

Builder Journal

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Week 1 – AI Summer Fellowship 2026

Computer Vision Engineering Track

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Why did you choose Computer Vision?

I chose Computer Vision because it is one of the most exciting fields of Artificial Intelligence. It enables computers to understand and analyze images and videos, making it possible to solve real-world problems through automation. I enjoy working on practical projects that involve image processing and visual data. Learning Computer Vision will help me build applications that can make a positive impact in healthcare, security, transportation, and many other industries.

What applications interest you the most?

The applications that interest me the most are object detection, medical image analysis, autonomous vehicles, and smart surveillance systems. I am particularly interested in developing intelligent systems that can detect and recognize objects accurately. In the future, I also want to explore facial recognition, gesture recognition, and AI-powered healthcare solutions using Computer Vision.

What did you learn this week?

During Week 1, I learned the fundamentals of Computer Vision and image processing. I studied how digital images are represented using pixels, the differences between RGB and grayscale images, and the importance of preprocessing techniques such as blurring, thresholding, and edge detection. I also learned how to use OpenCV for image processing tasks and Streamlit to develop an interactive Vision Toolkit application. Additionally, I improved my understanding of Git, GitHub, and project documentation.

Biggest challenge

The biggest challenge I faced was setting up the development environment and resolving library and interpreter issues in Visual Studio Code. Initially, my project showed multiple import errors even though the required packages were installed. I also faced issues with the webcam module and project configuration. These problems delayed development, but they helped me understand how Python environments and project structures work.

How did you solve it?

I solved these issues by creating and using a virtual environment, installing the required Python libraries, selecting the correct Python interpreter in Visual Studio Code, and fixing the webcam module. I also tested each feature of the Vision Toolkit individually to ensure that every function worked correctly. This experience improved my debugging and problem-solving skills.

Goals for Week 2

For Week 2, my goal is to build a complete object detection system using YOLO and OpenCV. I want to learn how to prepare datasets, annotate images, train deep learning models, evaluate their performance, and deploy the final application. I also aim to improve my GitHub portfolio by documenting every project professionally.

Reflection

Week 1 provided me with a strong foundation in Computer Vision engineering. Building the Vision Toolkit and performing OpenCV experiments gave me practical experience with image processing techniques. I also learned the importance of writing clean code, maintaining project documentation, and using GitHub professionally. I am confident that these skills will help me successfully complete the upcoming weeks of the fellowship.